

Uniform complexity of the pure-cell insertion construction

Complete proof; three independent mathematical reviews completed

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This note audits the dependence on the number of controlled moments in the fixed-number construction. It bounds the data that determine the *small* dice. Denominators of higher polynomial coefficients, Riesz solves, and the final large dice are deliberately not bounded.

Take the complexity parameter $P \geq 2$ to be an integer, replacing it by its ceiling when necessary. Call a pure-cell model of complexity at most P if the following quantities are at most P : the number of labels and cells, the common endpoint-grid denominator, the common denominator of all prescribed raw cell moments, polynomial degrees, and the maximum and reciprocal positive minimum of each active density piece. Ordered target degree and the number of prescribed low moments are included among these quantities. Each label has total mass one. An inactive density is zero and is excluded from the positive-minimum condition.

Lemma 1 (Uniform local parameter dependence). *There is an absolute constant C such that the synchronized local completion theorem can be carried out with all its controlled output data bounded by*

$$P^{(q+2)^C}.$$

Here the input family has at most P instances, degree and target order at most P , density bounds $P^{-1} \leq w_\alpha \leq P$, $q \leq P$, prescribed node counts $K_\alpha \leq P$, and the quantities $K_\alpha \int t^j w_\alpha$, $j \leq 2q+1$, have one common denominator at most P . The counts must be sufficiently large for the separated quadrature; a sufficient lower bound has the form

$$K_\alpha \geq e^{C(q+1)}(a^{-1} + b + 1)^C(m + \deg w_\alpha + q + 2)^4,$$

with a change of the absolute constant, where a, b are the actual common density bounds. Increasing the fixed polynomial degree in this last display would make no difference to any conclusion below.

Proof. We give explicit uniform types of all losses in `synchronized_local_moment_completion.tex`. This avoids treating its fixed- q constants as unspecified functions. For this audit, a bound is of the permitted type if its logarithm is at most $(q+2)^C \log P$ for some absolute C . A bounded number of products, fixed powers, or powers of order polynomial in q preserve this class after increasing C .

The separated weighted quantile/Newton lemma has only fixed powers of $a^{-1}, b, m + \deg w + q + 2$. For example $\|Q'\|_\infty \leq a^{-1}$ and $\|Q''\|_\infty \leq C(\deg w + 1)^2 b a^{-3}$ for the quantile map. The derivative Gram is bounded below by a , and the polynomial Markov bounds for its perturbation have fixed powers. Forcing $q+2$ control nodes costs a further polynomial in q : delete $q+2$ Gram columns and move that many nodes by $O(1/K)$. The required quadrature degree is at most $C3^q(m + \deg w + q + 2)$. These observations give the stated sufficient lower bound after enlarging C .

The universal feature conditioning is now quantitative in `uniform_prefix_conditioning.tex`. For perturbation size η , its inverse norm is at most

$$P^{C(q+1)^3} \eta^{-2q}.$$

The inverse Gram and the fixed Newton right inverse therefore cost at most another fixed power of this quantity, times fixed powers of P and q . Norms of all feature derivatives are bounded by fixed powers of their degrees, with factors at most polynomial in q . Thus, once η^{-1} is of the permitted type, sufficiently fine rational rank rounding preserves the fixed-inverse contraction with permitted precision. There is no product over the number of Newton iterations: one fixed right inverse and its contraction-radius bound control the complete solve.

For clarity, the remaining uniform losses are as follows.

1. Rescaled moment systems on a gap or box, with orders at most $q + 1$, have inverse norms bounded by $e^{C(q+1)^2 \log(q+2)} \Delta^{-C(q+1)^2}$, where Δ is the cell length. This follows directly from the triangular Legendre basis, binomial coefficients, and the nonzero diagonal moments.
2. The degree- q smoothed quotient estimate has inverse powers of the minimum gap of order $q+3/2$ and polynomial degree power $2q+4$. Its constants can be bounded by $e^{C(q+1)^2 \log(q+2)}$. Indeed its proof uses $q + 1$ derivatives, repeated integration, and binomial jet cancellation. The inverse Markov bound is $(CM^2/\Delta)^{q+1}$. The auxiliary backward recurrence over the polynomial degree is a geometric series with ratio at most $1/2$, not a product of degree-many constants.
3. The shared control-feature matrix has size $q + 2$. Its baseline is the rational Vandermonde matrix at $i/(q + 3)$, with rows $1, t, t^2/2, \dots, t^{q+1}/(q + 1)$. Its inverse norm is at most $e^{C(q+1)^2 \log(q+2)}$ by its determinant and cofactors. If its rounded entries have denominator L , a common denominator for its rational inverse is at most $(q+2)!L^{q+2}$, up to a fixed power of L ; all entries are bounded. Thus the same inverse used in every instance introduces a permitted common denominator.
4. The low Legendre modes for box moments cost at most $e^{C(q+1)^2 \log(q+2)} \epsilon^{-C(q+1)^2}$. The next mode, of degree $q + 1$, has leading response $\epsilon^{q+1} c_q$ times the already controlled feature-evaluation matrix, where $|c_q|^{-1} \leq e^{C(q+1) \log(q+2)}$. Taylor remainder bounds have degree losses of order $D^{O(q)}$ and powers of ϵ of order polynomial in q . Thus restoring higher moments preserves all low conditional moments at permitted precision.

The constants in the first two items can alternatively be obtained by expanding the explicit proofs in `higher_prefix_positive_completion.tex` and `../independent_directions/quantitative_smoothed_quotient.tex`; no matrix determinant of size comparable to m is used to bound a controlled denominator.

The gap lower bound before smoothing is an inverse fixed power of P . Choose $\eta = P^{-(q+2)^{C_1}}$ to meet the positive-completion bound with slack, then a common box width $\epsilon = P^{-(q+2)^{C_2}}$, with C_2 large enough for the conditioned smoothing and the box-measure error in the Riesz solve. Finally choose the feature-rounding, center-rounding, and low-moment rounding grids in that order with exponents $(q + 2)^{C_3}$, $(q + 2)^{C_4}$, $(q + 2)^{C_5}$, increasing these absolute exponents as needed. They can be chosen as integer powers of an integer at most $2P$, with the prescribed common arithmetic factors multiplied in. Those factors have permitted size: the input denominator, $(q + 1)!$, and the one shared control minor just discussed. This is a fixed number of choices, independent of m and q .

Every smallness requirement in the reviewed construction is a product of the losses just listed. Hence these finitely many exponents can all be dominated by one absolute exponent C . High-dimensional rational minimum-norm or Riesz solves may create large coefficient denominators, but their numerical norms are bounded by the displayed coercivity/quotient estimates. Their prescribed low moments and their already fixed endpoints do not change. Positivity, cell lengths, masses, and degrees therefore retain permitted bounds. This proves the lemma. $\square$

Lemma 2 (Uniform grid reset). *Under the complexity convention above, the uniform-grid moment reset with reserve q and ordered target degree at most P produces complexity at most $P^{(q+2)^C}$, for an absolute C .*

Proof. In `uniform_grid_moment_refinement.tex`, choose a grid refining the old endpoint grid by a factor at most $q + 1$, increasing it by another fixed power of P if needed. The translated adjacent-cell moment determinant is

$$Q^{-(q+1)(q+2)/2} \det \left(\int_j^{j+1} y^r dy \right)_{r,j=0}^q.$$

The fixed determinant is the Vandermonde determinant at $0, \dots, q$, hence a positive integer $\prod_{j=1}^q j!$. Entries have denominators dividing $(q+1)!Q^{q+1}$ and bounded numerators. The inverse norm and its common denominator are at most $e^{C(q+1)^2 \log(q+2)} Q^{C(q+1)^2}$. The low-moment realization and the higher-moment Riesz restoration cost at most the same type of bound times $(M+1)^{2q+3} Q^{q+3/2}$. Choose the one rounding denominator larger than their product, the original common low-moment denominator, and the reciprocal desired positivity margin. All factors have permitted size. The original density is retained and only these small corrections are made, so no coefficient denominator enters this rounding precision. This proves the claim. $\square$

Theorem 1 (A uniform count bound). *There is an absolute constant C such that, for all integers $n > k \geq 1$, a permutation-fair collection of n equiprobable finite dice exists in which k specified dice all have the same number S of faces satisfying*

$$S \leq (n + 2^k + 2)^{\exp(Ck^2)}.$$

The other $n - k$ dice may be taken equal-sized, with no asserted bound on that common size.

Proof. Use the reviewed induction in `./independent_directions/fixed_number_polynomial_dice.tex`, with reserves $q_j = 2^{k-j+1} - 1$ and ordered target degree $M = n + 2^k$. The initial one-uniform model has low moments $1/(a+1)$ through $q_1 = 2^k - 1$, whose common denominator divides $(2^k)!$. Thus its complexity is at most $P_1 \leq (n + 2^k + 2)^{C(2^k+1)}$. The extra factor $2^k + 1$ in $\log P_1$ is absorbed by the final $\exp(Ck^2)$ estimate.

Suppose the current pure uniform-grid model has complexity P_j and reserve $2q + 1$. On a cell $[A/Q, (A+1)/Q]$ of old mass γ , choose one global N divisible by the common raw-moment denominator, and put $K = N\gamma$. A choice $N \leq P_j^{C(q+2)}$, made sufficiently large, supplies the local separated quadratures in Lemma 1 for every cell. For every $a \leq 2q + 1$, the normalized local moment satisfies

$$K\mu_a = N \sum_{l=0}^a \binom{a}{l} Q^l (-A)^{a-l} \int_{\text{cell}} x^l d\nu(x) \in \mathbb{Z}.$$

Thus the local low-moment targets have common denominator one before the harmless factors $1/(s+1)$ in the feature targets. Apply Lemma 1 simultaneously to all cells with its shared control matrix and shared grids. After affine rescaling, common low moment denominators multiply only the common integers N, L, E, Q^q ; there is no least common multiple over cells. All these data have complexity $P_j^{(q+2)^C}$. Lemma 2 then restores one uniform pure grid, with the same form of bound after enlarging C . In particular,

$$\log P_{j+1} \leq (q_j + 2)^C \log P_j.$$

As $q_j + 2 \leq 2^{k-j+2}$, multiplying these factors over $j < k$ gives

$$\log P_k \leq \exp(C'k^2) \log(n + 2^k + 2).$$

Final separated quadrature on every pure cell, with one common global count and at least $(n - k)^2$ nodes per cluster, costs a fixed further power of P_k . The uniform quadrature proof has fixed polynomial losses in the numerical density bounds and degrees. It preserves all rational cluster moments and produces distinct interior nodes. The reviewed clustered lifting theorem then rationalizes these k finite special dice and corrects the other $n - k$ continuous uniforms. It preserves the special counts; its possibly enormous denominators affect only the final equal-sized old dice. This proves the stated bound on S . $\square$

Theorem 2 (A growing number of subexponential dice). *There is a constant $c > 0$ such that, with $k = \lfloor c\sqrt{\log n} \rfloor$, one may prescribe k dice in a fully permutation-fair n -die family to have the same count $S \leq \exp(n^{1/2})$ for all sufficiently large n . More generally, if $k(n) = o(\sqrt{\log n})$ and $k(n) \rightarrow \infty$, then the same construction gives $S \leq \exp(n^{o(1)})$.*

Proof. Choose c small enough that $Cc^2 < 1/3$ in Theorem 1. Then $2^k = n^{o(1)}$, and $\log S \leq n^{1/3+o(1)} \log n \leq n^{1/2}$. For the second statement, $\exp(Ck^2) = n^{o(1)}$ and $\log(n+2^k+2) = n^{o(1)}$. $\square$

The assertion controls only the specified dice. It is not an $\exp(O(n))$ bound for the total family and does not establish polynomial equalization. The absolute constants are deliberately unoptimized; the uniform dependence, rather than a useful numerical threshold, is the conclusion.